

Neural means and kernel corrections for an EMIT atmospheric radiative-transfer emulator

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Abstract

Imaging-spectroscopy retrievals call an atmospheric radiative-transfer model many times per spectrum, and at mission data rates a learned emulator of the tabulated physics is a practical necessity. We compare neural and kernel emulators, and their couplings, on a tabulated model over the 285-band EMIT wavelength grid under one protocol: ten random splits, every choice made on each split's own validation rows, one test read per family, and every number reported with its spread across splits. The map is smooth, six-dimensional and low rank, and an exact Matérn regression fitted on all 18,884 training rows with one length scale per input is the strongest single model at 0.095% relative radiance error, against 0.76% for a cubic polynomial and 0.39% for a fully connected network; its fitted metric recovers the physics, giving the relative azimuth a sixteen- to thirty-two-fold longer length scale on the transmittances. A kernel regression of the network's residuals brings the network to 0.14% and a per-component convex stack reaches 0.087%, while per-coordinate selection does not transfer from validation to test. Learning curves from 500 rows to the full block put every family except the cubic at close to $n^{-1/2}$ on the three smooth components and at zero on the spherical albedo, and the corrected network below both of its parents at every split from 2,000 rows on; an identity for the residual correction, whose error is the kernel regression's residual on the network's residual, says why its curve takes the network's exponent and the kernel's level, and why the path radiance is the exception. The flattening of the kernel's curve at the largest size is an artefact of choosing its hyperparameters on a subsample of the solve, which over-smooths; tuning on every row removes it. A metric read off a cubic surrogate's gradients, with its exponent chosen on validation, closes four fifths of the distance from the isotropic kernel to the searched one for one cubic fit and six finite differences. The kernel is indifferent to the output rank above 32 coefficients while the correction has an optimum near it. A network four times wider is a worse emulator and a better representation: an exact kernel on its features beats the per-input kernel at every split, and a convex stack built on it is the lowest row of the study at 0.078%. We also measure what the reflectance metric returns for exact inputs: in the deep absorption windows the transmitted flux approaches zero, the inversion is ill-conditioned and at exactly zero flux non-identifiable, so the exact tabulated components put through it already give a root-mean-square error of half a reflectance percentage point. That number describes the evaluation at those entries; it is not a lower bound that every emulator must pay. On the OCO-2 problem of the companion paper, where the input is higher-dimensional and the map rougher, none of the constructions that win here transfers, and the kernel pays only inside the network's representation.

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1 Introduction

Imaging spectroscopy missions such as EMIT [13], SBG, and AVIRIS-NG invert measured radiances for surface and atmospheric quantities, and the inversion loop calls an atmospheric radiative-transfer model many times per spectrum. At mission data rates the physics code itself (MODTRAN in the pipelines relevant here [1]) is far too slow, so operational retrievals interpolate precomputed lookup tables, and a body of work replaces table and physics alike with learned emulators: Gaussian-process surrogates [5], the neural-network MODTRAN emulator that serves the EMIT mission’s operational atmospheric correction [3, 2], and, recently, physics-guided and multi-fidelity architectures for correction coefficients [4]; network training itself can draw on a growing toolbox of refinements [7, 8]. Both families have a record on emulation tasks of this kind: networks learn their own representations and scale easily to wide spectral outputs [10, 11], while kernel and Gaussian-process methods come with exact solves and a strong record both on operator-learning benchmarks [9] and on smooth, low-dimensional regression [14]. Which family is preferable for a given emulation problem, and whether they can be combined so that neither’s advantage is given up, are practical questions that the structure of the data itself should decide.

This paper studies those questions on a tabulated radiative-transfer emulation problem over the EMIT wavelength grid, using the coupling construction of [6]: train a neural network as the mean in the metric the application reports, correct it with exact Matérn kernel regressions — of its residuals on the input space, and of the target on the network’s own learned features — and let a per-coordinate selection on held-out data choose, coefficient by coefficient, among the network, the kernels, and the corrected predictors. We ask what structure the dataset has, which model families that structure favors, and whether the kernel correction of a neural mean improves on both of its parents here.

The dataset answers the first question quickly (Section 2). The input is six-dimensional, and two coordinates carry most of the variance of the output; the output spectra are smooth curves whose empirical rank is a few dozen out of 285 bands; and the map from state to spectrum is smooth enough that cubic polynomial regression already lands near two percent relative error. In the taxonomy of [6] this is the regime that favors exact kernel methods, in contrast to emulation problems with higher-dimensional reduced states where representation learning carries more of the weight. The interest of the comparison is therefore not whether some exotic architecture wins, but how the familiar families divide the problem between them when the geometry is easy, and whether the coupling machinery preserves the best of each.

Sections 3 and 4 give the protocol and the measurements. The results are these. With every training row in the solve and a length scale per input, the exact kernel is the strongest single model on this table, and its learned metric recovers the physics of the components. The kernel regression of the network’s residuals does what it was built to do, handing the network the kernel’s accuracy on the smooth components, but it is no longer the lowest row. A per-component convex stack of the heads is, by about one standard deviation across the ten splits, while per-coordinate selection does not transfer from validation to test. The reflectance metric has a difficulty of its own, set by the atmospheric absorption windows in which the transmitted flux approaches zero: there the sensitivity of the inversion to the path radiance and the two flux components is unbounded, and at exactly zero flux the physical reflectance is not identifiable at all. We measure what the metric returns for exact inputs and summarize reflectance through a statistic the divergence cannot dominate. Section 5 adds what a single training size cannot show: learning curves from 500 rows to the full block, refits at other output ranks, a ladder of

network widths, and the two questions those curves raise — where the kernel’s hyperparameters should be chosen, since the protocol used here turns out to cost it a measurable amount, and whether its metric can be read off the data instead of searched for. Section 6 gives the identity behind the residual correction, which says what the corrected network can inherit from each of its parents and accounts for the shape of every curve. Section 7 records where the constructions that win here do not transfer, on the OCO-2 problem of the companion paper [6], and Section 8 applies the same family set to ten further emulation configurations.

2 Data and protocol

The data are $n = 23,313$ tabulated radiative-transfer evaluations. Each input $x \in \mathbb{R}^6$ collects the atmospheric and viewing state; the coordinate ranges are consistent with the standard lookup-table axes for imaging-spectroscopy atmospheric correction [2] — an aerosol optical depth in $[0.05, 1.2]$, a surface elevation in $[-0.5, 6]$ km, a water-vapor column in $[0.05, 7]$ cm, a relative azimuth in $[0, 180]^\circ$, a solar zenith angle in $[0, 65]^\circ$, and a view zenith angle within 30° of nadir. Each output is a set of four spectral components on the same 285-point EMIT wavelength grid spanning 381–2493 nm: path radiance Y_1 , direct flux Y_2 , diffuse flux Y_3 , and spherical albedo Y_4 (Figure 1). The quantities retrieved downstream are the radiance at a test surface reflectance ρ ,

$$L(\rho) = Y_1 + \rho \frac{Y_2 + Y_3}{1 - \rho Y_4}, \quad (1)$$

and the reflectance recovered by inverting (1) with predicted components at the true radiance,

$$\hat{\rho} = \frac{L - \hat{Y}_1}{\hat{Y}_2 + \hat{Y}_3 + \hat{Y}_4(L - \hat{Y}_1)}, \quad (2)$$

both evaluated at $\rho = 0.7$.

The protocol is the same for every family. The table is split at random into a test block of 2,331 evaluations and a training block of 20,982, and a further ten percent of the training block, 2,098 points, is set aside for validation, leaving 18,884 training points. The split is drawn ten times, from ten seeds, and every model is trained, selected and scored once per seed: inputs and outputs are standardized on the training rows of that seed, every hyperparameter and every combination weight is chosen on that seed’s validation points, and the test block of that seed is read once per family at the end. The numbers reported below are means and standard deviations over the ten seeds, so a difference between two rows can be read against the spread of each. An earlier version of this study used a single split and consulted its test block during exploration; none of its numbers is reused here, and the ten seeds were drawn afresh, by a plain permutation rather than the earlier library routine, so no code path shares randomness with that split. The validation rows are used for early stopping, checkpoint selection, hyperparameter selection and the combiners, and are therefore not an independent calibration split; no coverage guarantee is claimed from them.

Three structural facts, measured before any model was chosen, organize everything that follows.

The outputs are low rank. On standardized training outputs, 99.99% of the variance of every component lies in the first 11–25 principal components, and 99.9999% in at most 44 (Figure 2, left); the minimum adjacent-band correlation across the four components is 0.994952 on the training rows. All regressors below therefore act on the first 64 principal coefficients —

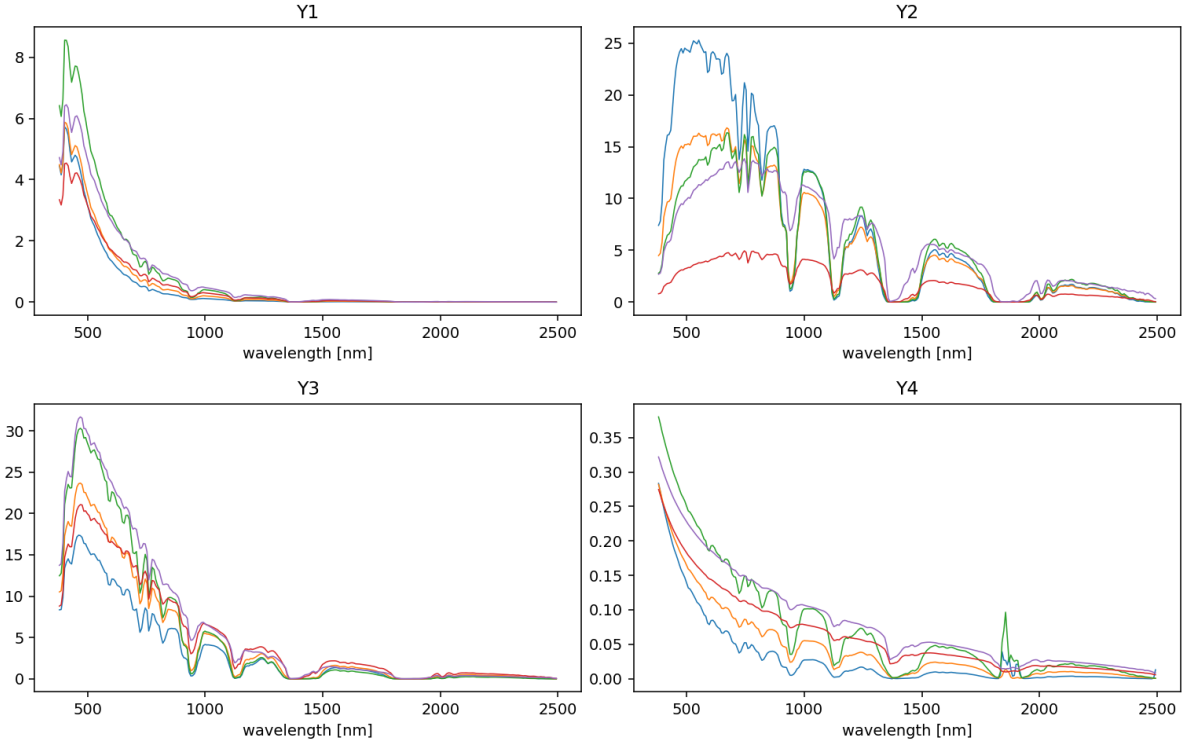


Figure 1: Example output curves across the EMIT wavelength grid, five samples per panel. The curves are smooth, share the deep water-vapor absorption features near 1380 and 1900 nm, and differ mainly by smooth rescalings along the state axes.

a lossless reduction at the 10^{-6} level, verified by round trip — except the networks, which we also train directly on the 285 standardized bands.

The measured input relevance is strongly anisotropic. Shuffling the aerosol coordinate destroys a linear model’s validation R^2 (a drop of 0.89); the solar zenith follows at 0.54; elevation and water vapor sit near 0.1; the two remaining angles barely register (Figure 2, right). The values reported for these probes, and for the other structure measurements used to motivate architectures, are their validation-split rescaling; the initial exploratory versions were test-scored as disclosed above. Six nominal dimensions have four coordinates with measurable first-order relevance, two of them dominant. This finite-sample diagnostic helps explain why smooth methods work, but a selected scalar length scale does not remove anisotropy. Any theorem attributing generalization to anisotropic metric adaptation would also have to control regularization bias, selection of scale and nugget, and the empirical output-rank truncation. We use anisotropy as a measured property of this split, not as such a guarantee.

The map is smooth. A linear model in the standardized state reaches $R^2 = 0.83$; distance-weighted 5-nearest-neighbors does worse (0.74), which already says local averaging is the wrong bias here; and a cubic polynomial ridge reaches mean relative L^2 errors near two percent (Table 1). Smooth, low-dimensional, low-rank: in the taxonomy of the companion paper this is the regime of the Helmholtz and Navier–Stokes benchmarks, where the kernel stage won the internal selection, not the regime of its OCO-2 problem, where representation learning carried the problem.

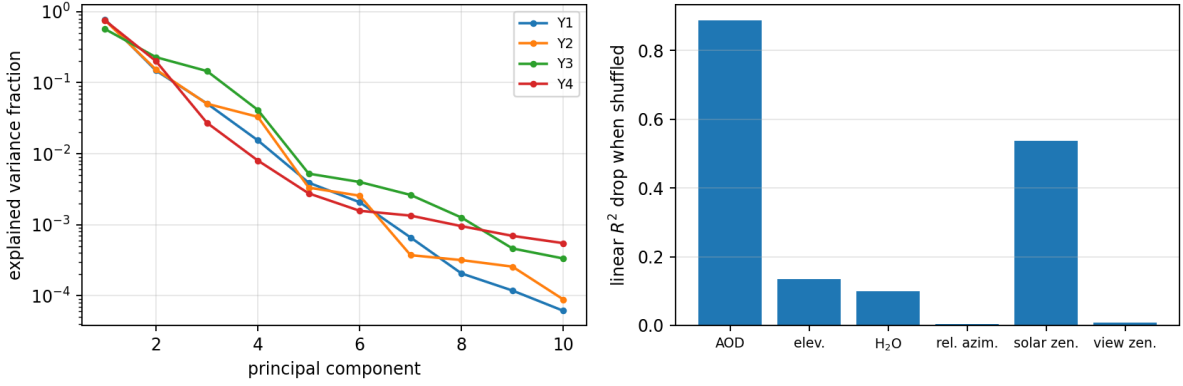


Figure 2: Left: explained-variance spectra of the standardized outputs. Every component concentrates 99.99% of its variance in 11–25 principal components out of 285 bands. Right: drop in linear validation R^2 when a single standardized input coordinate is shuffled. Aerosol optical depth and solar zenith angle dominate; the azimuth and view-zenith coordinates are almost inert at first order.

One more property of the task matters for reading any reflectance number. In the deep absorption windows the transmitted flux $Y_2 + Y_3$ collapses toward zero — in this table, to values as small as 10^{-21} — and the inversion (2) is undefined at a fully opaque band (0/0): the physical reflectance is non-identifiable there. The protocol maps a non-finite inverse to zero, but that branch is not what creates the measured true-component error. The exact components, put through the same evaluation, give a reflectance RMSE of 0.0054 against the true $\rho = 0.7$, produced entirely by 0.25% of (sample, band) entries — 1,652 of 664,620, measured on the earlier single split, whose test block held one row more than the ten drawn here — whose denominators cancel to below 2×10^{-13} ; every value returned is finite (the protocol’s zero-fill for non-finite inversions is never triggered by the true components), and the effect is invisible in component errors. What this measures is the conditioning of the inversion at those entries and not the accuracy of any emulator: it is what the evaluation returns when the components handed to it are exact. It is not a lower bound on an emulator’s reflectance error. A predictor whose component errors offset the cancellation would score below it, and a predictor with generic errors of its own can score far above it; what the measurement establishes is that a mean-square reflectance summary on this table reports the conditioning of these entries alongside whatever the emulator does, which is why reflectance is summarized below by a statistic the tail cannot reach.

Differentiating (2) separates the components. Write $D = \hat{Y}_2 + \hat{Y}_3 + \hat{Y}_4(L - \hat{Y}_1)$ for its denominator. Then

$$\frac{\partial \hat{\rho}}{\partial \hat{Y}_4} = -\hat{\rho}^2, \quad \frac{\partial \hat{\rho}}{\partial \hat{Y}_2} = \frac{\partial \hat{\rho}}{\partial \hat{Y}_3} = -\frac{\hat{\rho}}{D}, \quad \frac{\partial \hat{\rho}}{\partial \hat{Y}_1} = -\frac{\hat{Y}_2 + \hat{Y}_3}{D^2}. \quad (3)$$

The sensitivity to the spherical albedo is bounded and of order one — about 0.49 at $\rho = 0.7$ — at every band, so an albedo error moves the recovered reflectance by about half its own size everywhere, in the absorption windows and outside them alike. The divergence as the transmitted flux vanishes belongs to the path radiance and the two flux components, whose sensitivities carry D^{-1} or D^{-2} . The ill-conditioning is continuous rather than confined to those

entries: for the strongest classical model of the earlier single-split run, 7.4% of test entries land more than 0.5 away from the true reflectance, and 95% of those lie at bands whose true transmitted flux is below one percent of its median value. Reflectance is therefore summarized in what follows by its median absolute error, which the absorption windows cannot reach; outside the cancellation tail a perfect emulator scores zero to within 10^{-8} .

3 Models

All models are trained per component and evaluated in physical units after undoing the standardization; unless stated otherwise they regress the 64 leading principal coefficients of the standardized component, a reduction lossless at the 10^{-6} level (Section 2). Every hyperparameter named below is selected on the validation set of the seed at hand. Everything runs on CPU.

Reference family. Cubic polynomial ridge regression, acting on all monomials of degree at most three in the six standardized inputs, with the ridge weight selected from $\{10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$. It is the benchmark of smoothness: a map that a cubic fits to two percent is a smooth map.

Exact kernel ridge regression. Three constructions, all Matérn kernels on the standardized inputs with the length scale expressed as a multiple of the median pairwise distance of the fit set. The first is the construction of the earlier study, kept for comparison: a Matérn-5/2 kernel fitted on a uniformly drawn subset of 4,000 training points, one eigendecomposition per length scale, multiplier in $\{1, 2\}$, nugget in $\{10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$. The second is the exact solve on every training row: smoothness $\nu \in \{3/2, 5/2\}$, multiplier in $\{0.5, 1, 2, 4\}$ and the same nugget grid, tuned on validation from a 6,000-row training subsample and the winner refit on all 18,884 rows by one Cholesky factorization. The third gives each input its own length scale: starting from the isotropic winner, a coordinate search over per-input multipliers in $\{1/4, 1/2, 1, 2, 4\}$, two sweeps, scored on validation, after which the smoothness, scale and nugget are re-tuned at the found metric and the solve is refit on every row. The input relevance measured in Section 2 spans more than an order of magnitude across the six coordinates, which is what the third construction is for.

Fully connected networks. Three ELU hidden layers of 512 units per component, trained on the 285 standardized bands with AdamW at learning rate 10^{-3} , cosine decay, batch size 1024, weight decay 10^{-6} , up to 150 epochs with early stopping on validation loss (patience 25). Five networks are trained per component and seed from different initializations; the table reports the first and the mean of the five.

Kernel corrections of the neural mean. The construction of [6], with the exact stages now fitted on every training row. (i) *Residual correction*: an exact Matérn regression, on the standardized input space, of the network’s training residuals in coefficient space, added back to the network, its smoothness, scale and nugget selected by the validation error of the corrected prediction; applied to the single network and to the five-network mean. (ii) *Kernel on features*: the same exact regression with the input metric replaced by the network’s last hidden layer, standardized, so that the kernel’s exactness is spent in coordinates the network

model	component rel. L^2 [%]				radiance	reflectance
	Y_1	Y_2	Y_3	Y_4	rel. L^2 [%]	med. $ \hat{\rho} - \rho $ [%]
Cubic ridge	1.16±0.02	2.18±0.04	1.11±0.03	2.98±0.06	0.760±0.011	0.471±0.010
Matérn KRR, 4000-point fit	0.17±0.01	0.61±0.04	0.29±0.02	2.44±0.87	0.332±0.029	0.178±0.030
Matérn KRR, exact on all rows	0.08±0.01	0.42±0.03	0.20±0.00	2.62±0.18	0.239±0.016	0.141±0.018
Matérn KRR, per-input scales	0.08±0.01	0.16±0.02	0.08±0.01	2.18±0.16	0.095±0.009	0.036±0.004
FC-DNN (3×512)	0.48±0.02	0.97±0.04	0.41±0.02	3.02±0.18	0.387±0.012	0.245±0.008
Five-network mean	0.37±0.01	0.67±0.03	0.28±0.01	2.51±0.09	0.251±0.006	0.163±0.003
DNN + residual KRR	0.10±0.00	0.30±0.02	0.16±0.01	2.43±0.08	0.141±0.006	0.089±0.008
Ensemble + residual KRR	0.08±0.00	0.27±0.03	0.13±0.00	2.42±0.17	0.123±0.007	0.080±0.008
Kernel on network features	0.18±0.01	0.17±0.01	0.10±0.01	1.99±0.08	0.098±0.007	0.054±0.004
Per-coordinate selection	0.19±0.01	0.44±0.10	0.13±0.02	2.25±0.18	0.193±0.022	0.123±0.023
Convex stack	0.07±0.00	0.16±0.01	0.09±0.00	1.94±0.06	0.087±0.006	0.048±0.003

Table 1: Test errors under the protocol of Section 2, mean $\pm$ standard deviation over the ten splits. Component columns are mean relative L^2 errors; the radiance column evaluates (1) at $\rho = 0.7$; the last column is the median absolute reflectance error of the inversion (2), in reflectance percentage points, under the zero-fill evaluation of Section 2. The median is used because the sensitivity of (2) to the path radiance and the two flux components diverges continuously as the transmitted flux vanishes, so mean-square and tail-percentile summaries are dominated, for every model alike, by near-opaque entries; the exact components put through the same evaluation give an RMSE of about half a percentage point there (Section 2), which is a property of those entries and not a bound on any emulator. The tail summaries are stored with the per-seed records.

has already straightened. This is the post-hoc counterpart of deep kernel learning [24], in which the representation and the kernel are trained together, and of the kernel-flow regularization of a network’s inner layers [12]; here the representation is fixed first and the kernel is solved exactly on it. (iii) *Per-coordinate selection*: for each of the 64 output coefficients independently, pick on validation among the exact kernel, the network, the ensemble, the two corrected predictors and the feature kernel, and assemble the test prediction from the winners. (iv) *Convex stack*: per component, a convex combination of the same six heads with weights fitted on the validation points in the physical relative metric, in the manner of stacked generalization [22] and stacked regression [23]. The choice of weighting rule for pools of this kind, and its relation to constrained minimum-variance weighting, is studied separately and is not part of this comparison.

What is evaluated at other budgets. Section 5 repeats the isotropic exact kernel, the network, its residual correction and the feature kernel at nested training sizes, at other output ranks and at a wider network; the table of Section 4 is the full-size, rank-64, width-512 cell of that grid.

4 Results

Table 1 collects the test errors. Every entry is a mean over the ten splits with its standard deviation, and the spreads are small enough that most of the orderings discussed below hold seed by seed; where they do not, the text says so.

The scale of the problem. The cubic ridge reaches 1.86% mean component error and 0.76% on the radiance: the map is smooth, and a polynomial in six variables recovers it to two percent. The earlier construction of the kernel regression, a Matérn-5/2 on a 4,000-point subset, divides both numbers by more than two at seven of the ten splits (0.74% and 0.31%); at the other three its albedo fit collapses (3.3–4.3% against 1.9%), which is what a fit capped at a random subset can do, and its ten-split mean (0.88%, 0.33%) carries that fragility.

Exact kernels. Fitting the same kernel on every training row, one Cholesky factorization of the $18,884 \times 18,884$ Gram matrix per component, lowers the radiance error to 0.24% and the path-radiance error by half (0.08% against 0.17%), and it has no collapsing split (2.3–2.9% on the albedo at every seed), but where the capped fit is sound it is worse on the albedo (2.6% against 1.9%), so the mean component error moves the wrong way at those splits (0.83% against 0.74%). The albedo is the component whose principal spectrum decays slowest and whose values sit within a few percent of zero over much of the grid, and it is also the one where the transfer of a scale and nugget tuned on a 6,000-row subsample to the full solve is least reliable. A length scale per input removes most of this: the per-input kernel reaches 0.63% on the components and 0.095% on the radiance, with the best reflectance column of the table (0.036 percentage points), and its albedo column (2.2%) sits between the two isotropic solves. The learned metric is the physics of the components: for the two flux components and the spherical albedo the coordinate search gives the relative azimuth a length scale sixteen to thirty-two times longer than the others — a median of sixteen on the direct flux and the albedo and of thirty-two on the diffuse flux, at every one of the ten seeds for the two flux components and at nine of the ten for the albedo, whose search leaves the azimuth unscaled at the tenth — and the view zenith two to sixteen times longer, while for the path radiance, which does depend on the viewing geometry, the metric stays nearly isotropic. Transmittances and the spherical albedo do not depend on azimuth, and the kernel finds that out from the validation rows alone. On this table the per-input exact kernel and the kernel on network features (below) are, within the seed-to-seed spread, the strongest single models.

Networks. The fully connected network lands at 1.22% and 0.39%; the mean of five independently initialized networks at 0.96% and 0.25%, a third off the radiance error for five times the training cost, the usual decorrelation gain. Neither reaches the isotropic exact kernel on any of the first three components, and the network’s advantage on the albedo that the single-split study reported does not survive replication: on Y_4 the five-network mean (2.5%) is behind the 4,000-point kernel wherever that kernel’s fit is sound (1.9% at seven of the ten seeds) and behind the exact kernel with per-input scales at every seed.

Kernel corrections of the neural mean. An exact Matérn regression of the network’s residuals on the standardized inputs takes the single network from 0.39% to 0.14% on the radiance and the five-network mean from 0.25% to 0.12%; the kernel fitted in the network’s last-layer features (DKR) reaches 0.61% and 0.098%, indistinguishable from the per-input kernel on the components and on the radiance — 0.098 against 0.095, the lower at six seeds of ten — but half again worse on the reflectance median, 0.054 against 0.036 points, and worse there at every one of the ten. The correction therefore does exactly what it did on the single split — it hands the network the kernel’s accuracy on the smooth components while keeping the network’s own representation — but with the kernel now allowed its own input metric the corrected network

is no longer the lowest row: the ordering by radiance error is stack, per-input kernel, DKR, corrected ensemble, corrected network, with the first four within 0.04 percentage points of one another.

Selection and stacking. Both combiners here draw on the same six heads — the isotropic exact kernel, the network, the five-network mean, the two corrected predictors and the feature kernel — and not on the per-input kernel, which is the strongest single row of the table. The per-coordinate selection among those six, made on validation, does not transfer: its test radiance error (0.19%) is worse than three of the heads it selects among, and its spread over seeds (± 0.02) is among the largest in the table. The per-component convex stack, with weights fitted on the validation rows in the physical relative metric, is the best aggregate row (0.57% components, 0.087% radiance, 0.048 reflectance points) and wins the path radiance and the albedo columns outright; its gain over the strongest single row of the table is eight percent on the radiance and ten percent on the components, of the order of one standard deviation of either; measured against the best model the stack actually contains, the kernel on network features, it is eleven percent on the radiance and seven percent on the components. The paired view is sharper than the marginal one: split by split, the stack beats the per-input kernel on the components in nine of the ten splits (mean difference 0.06 points, spread 0.05) and beats DKR in all ten (0.047 ± 0.006), while the per-input kernel and DKR trade places (six of ten). The stack is the best row, and the residual correlations among the members it actually contains say why it can be: on the path radiance the residuals of the isotropic exact kernel and of the five-network mean are correlated at 0.27 ± 0.09 , below the ratio of their errors (0.34) at every one of the ten splits, which is the classical admission condition for a combination to improve on its better member [25], in the form used in [6]. Not every pair in the pool admits: the two residual-corrected predictors are correlated at 0.88 on the path radiance and at 0.99 on the albedo, and are largely two views of the same predictor. The heads are complementary in the sense that matters for a stack — the kernel is best on the smooth components, the corrected networks on the albedo — and a per-coordinate winner-take-all rule is too brittle a way of using that complementarity when the validation and test blocks are drawn from the same table but the picks are made on 2,098 rows.

Comparison with the single-split study. The earlier version of this study reported the corrected network as the lowest row, at 0.25% radiance error, and the exact kernel on the raw state at 0.41%; both numbers came from a single split whose test block had been consulted during exploration and from a kernel fitted on 4,000 points with one length scale. Under the fresh-seed protocol the corrected network reads 0.14% and the exact kernel with per-input scales 0.095%. The smoothness of the map, the complementarity of the heads across components and the mechanism of the residual correction hold under replication; the claim that the corrected network is the best model on this table does not.

The albedo. Every family is worst on Y_4 by an order of magnitude, and the per-band errors show why: the spherical albedo’s values in the short-wave infrared are close to zero, so relative error is large for every model there, and the component is hard for reasons of signal-to-noise rather than of model class. The stack’s 1.94% is the best number any construction reaches at this training size.

training rows	seeds	Cubic ridge	KRR, isotropic	KRR, per-input	FC-DNN	DNN + residual KRR	kernel on features
500	10	0.846±0.031	0.639±0.022	0.415±0.024	1.961±0.057	0.815±0.035	0.456±0.020
1,000	10	0.802±0.024	0.519±0.034	0.260±0.066	1.862±0.015	0.590±0.027	0.316±0.040
2,000	10	0.780±0.016	0.424±0.051	0.217±0.052	1.232±0.034	0.376±0.041	0.240±0.049
4,000	10	0.769±0.010	0.328±0.028	0.154±0.037	0.830±0.019	0.256±0.021	0.164±0.033
8,000	10	0.763±0.010	0.279±0.020	0.119±0.027	0.584±0.020	0.190±0.014	0.121±0.017
18,884	10	0.760±0.011	0.239±0.016	0.095±0.009	0.387±0.012	0.141±0.006	0.098±0.007

Table 2: Learning curves: radiance relative L^2 error [%] against the number of training rows, mean $\pm$ standard deviation over the ten splits. Each rung uses the first n rows of the seed’s training block; the validation and test blocks are those of Table 1, so the top rung is that table. The per-input kernel repeats its coordinate search at every rung. Component columns are in Figure 3 and in the per-seed records.

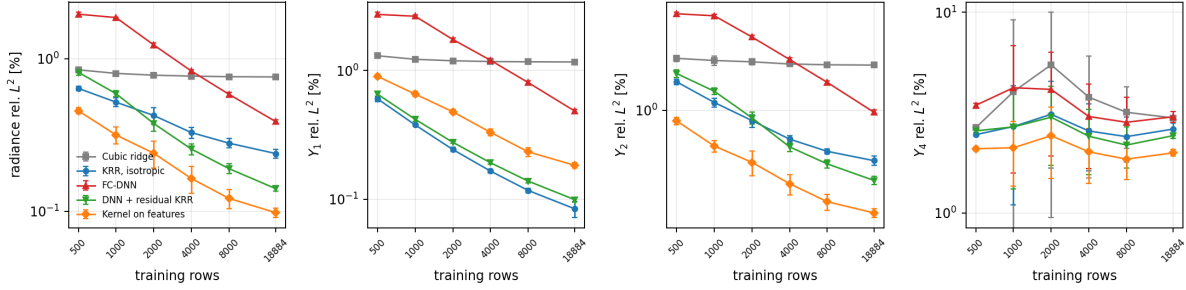


Figure 3: Learning curves on log–log axes for the radiance and for three of the four components, mean and standard deviation over ten splits. The cubic ridge is flat from the first rung; the kernel, the network, the corrected network and the feature kernel fall at close to $n^{-1/2}$ on the smooth components and not at all on the spherical albedo. The albedo’s spread at 1,000 and 2,000 rows comes from single splits whose albedo fit collapses, the failure Section 4 describes for a kernel capped at a random subset; the medians across splits are flat.

5 Learning curves, output rank and network width

Table 1 is one cell of a larger design: one training size, one output rank, one network width. This section varies each in turn, at the same ten seeds, with the same validation-based selection and one test read per family, for the isotropic exact kernel, the network, its residual correction and the kernel on its features; the cubic ridge is carried as the smoothness reference. The per-input kernel and the ensembles are run at full size only.

Learning curves. Each rung uses the first n rows of the seed’s training block, $n = 500, 1,000, 2,000, 4,000, 8,000$ and the full 18,884, with the validation and test blocks unchanged, so the rungs of one seed are nested and the full block is the top rung (Table 2, Figure 3). Three things are visible. First, the cubic is flat: from 0.85% to 0.76% over a factor of 38 in data, a bias floor reached by a thousand rows. Second, on the three smooth components every other family decays at close to $n^{-1/2}$ (Table 3): the least-squares slopes of log error against log n over the six rungs lie between -0.38 and -0.61 for the kernel, the network, the corrected network and the feature kernel on Y_1 , Y_2 and Y_3 , while on the albedo every slope lies between -0.09 and $+0.01$: the component whose values sit near zero over much of the grid is not data-limited at any size in this range, which is what the per-band anatomy of Section 4 suggested.

family	slope over the six rungs					slope of the last step				
	Y_1	Y_2	Y_3	Y_4	radiance	Y_1	Y_2	Y_3	Y_4	radiance
Cubic ridge	-0.03	-0.03	-0.03	-0.02	-0.03	-0.00	-0.01	-0.00	-0.08	-0.01
KRR, isotropic	-0.55	-0.38	-0.40	-0.01	-0.28	-0.38	-0.19	-0.17	0.10	-0.18
KRR, per-input	-0.45	-0.43	-0.43	0.01	-0.40	-0.32	-0.26	-0.29	0.12	-0.26
FC-DNN	-0.50	-0.49	-0.57	-0.09	-0.48	-0.59	-0.60	-0.55	0.07	-0.48
DNN + residual KRR	-0.52	-0.53	-0.61	-0.04	-0.50	-0.39	-0.33	-0.39	0.12	-0.35
Kernel on features	-0.45	-0.44	-0.46	-0.03	-0.43	-0.28	-0.22	-0.31	0.08	-0.25

Table 3: Learning-curve exponents: the least-squares slope of log error against log n over the six rungs of Table 2, and the slope of the last step alone, from 8,000 to 18,884 rows, per family and component.

For regression of a target in the Sobolev class H^s on a six-dimensional domain the rate of the root-mean-square error is $n^{-s/(2s+6)}$, which approaches $n^{-1/2}$ as s grows [16, 17]; the table is noiseless, so faster decay is not excluded by that theory, but slopes between -0.4 and -0.6 are what a target smooth enough that dimension no longer sets the rate looks like. Third, the kernel’s curve bends at the top. Its slope over the first five rungs is -0.59 , -0.44 and -0.47 on the smooth components, and its last step, from 8,000 to 18,884 rows, is -0.38 , -0.19 and -0.17 , while the network’s last step is as steep as its average (-0.59 , -0.60 and -0.55 ; Table 3). Part of this is the tuning protocol: the kernel’s smoothness, scale and nugget are chosen on a 6,000-row subsample at every rung, so above 6,000 rows the hyperparameters are those of a smaller problem, and Section 5.1 measures what tuning with every row in the solve recovers: all of it on the radiance, most of it on the components. The consequence for the comparison is that the ratio of the network’s radiance error to the kernel’s falls from 3.1 at 500 rows to 1.6 at the full block, or to 1.8 with the kernel tuned on every row. The kernel’s margin over the network on this table is a margin at this training size; whether it would survive ten times the data depends on where the kernel’s curve settles, which the bend leaves open.

Fourth, and this is the one a single training size cannot show: against the isotropic kernel the per-input metric buys a rate and not a constant. Its coordinate search, repeated at every rung, gives a slope of -0.399 on the radiance against the isotropic kernel’s -0.279 , and the ratio between them falls from 0.65 at 500 rows to 0.40 at the full block, with one step against that direction (0.50 at 1,000 rows, 0.51 at 2,000) and a fall at every other. Learning six numbers on validation is therefore not a fixed discount on a smooth problem: it is worth more the more data there is to spend on the coordinates, which is what removing an irrelevant direction should do, since the effective dimension left behind is what sets the rate. Against the kernel on network features the margin closes as the data grows. The per-input metric has the lower mean at all six rungs, but the split-by-split count runs 10, 9, 10, 7, 5 and 6 wins of ten from 500 rows to the full block, and the gap itself narrows from 0.415 against 0.456 at 500 rows to 0.095 against 0.098 at the top; the feature kernel’s own slope, -0.435 , is the steeper of the two. A metric searched on six validated numbers is the better buy on small data, and a representation learned from the data catches it up. On the albedo the per-input slope is $+0.01$: the one component no metric rescues.

The corrected network is below both of its parents on the radiance at every one of the ten splits from 2,000 rows on, and at no split at 500 or 1,000 rows, where its error sits between the network’s and the kernel’s. On Y_2 and Y_3 it is below both at every split at the two largest sizes; on Y_1 it never is, the isotropic kernel alone being the better head at every size (0.084%

PCA rank	variance outside the basis	KRR, isotropic	FC-DNN	DNN + residual KRR	kernel on features
16	7.5×10^{-4}	0.270 $\pm$ 0.015	0.390 $\pm$ 0.015	0.175 $\pm$ 0.007	0.148 $\pm$ 0.009
32	1.5×10^{-5}	0.243 $\pm$ 0.018	0.390 $\pm$ 0.015	0.139 $\pm$ 0.006	0.102 $\pm$ 0.009
64	8.2×10^{-8}	0.242 $\pm$ 0.018	0.390 $\pm$ 0.015	0.143 $\pm$ 0.006	0.100 $\pm$ 0.009
128	5.4×10^{-9}	0.242 $\pm$ 0.018	0.390 $\pm$ 0.015	0.150 $\pm$ 0.005	0.100 $\pm$ 0.009
285	0	0.242 $\pm$ 0.018	0.390 $\pm$ 0.015	0.172 $\pm$ 0.004	0.100 $\pm$ 0.009

Table 4: Output-rank ablation: radiance relative L^2 error [%] when the kernels and corrections regress the leading r principal coefficients, mean $\pm$ standard deviation over five splits. The second column is the fraction of the standardized variance the basis leaves out for the slowest-decaying component, the spherical albedo; at rank 285 it is zero. The network regresses the 285 bands directly and is the same predictor in every row.

against 0.099% at the full block). Its exponent is close to the network’s and its level is set by the kernel’s. Section 6 says why, and why Y_1 is the exception.

Output rank. The kernels and corrections regress the leading r principal coefficients of the standardized component; the network regresses the 285 bands and is the same predictor at every rank. Table 4 refits at $r = 16, 32, 128$ and 285 at five of the seeds beside the rank-64 rows of those seeds. The kernel is indifferent to the rank above 32: its radiance error reads 0.243, 0.242, 0.242 and 0.242 at 32, 64, 128 and 285, and 0.270 at 16, where the truncation leaves 7.5×10^{-4} of the albedo’s variance outside the basis. The kernel on features behaves the same way. The residual correction does not: it is best at 32 (0.139), a little worse at 64 (0.143), and worse again at 128 and 285 (0.150 and 0.172). The difference is in what the two regressions fit. The kernel regresses the component itself, whose tail coefficients are small and smooth functions of the state, and carrying them costs nothing. The correction regresses the network’s residual, whose tail coefficients are the network’s own errors along directions that hold 10^{-6} of the variance, and a kernel regression of those directions adds a term the 2,098 validation rows select but the test block does not reward. A rank at the point where the retained variance stops moving, 32 to 64 on this table, is the right setting for the correction, and the rank-64 setting of Table 1 is on the safe side of it.

A wider network. A network of three hidden layers of 2,000 units, trained for up to 500 epochs under the same optimizer, learning rate and early stopping as the 512-unit network, was run at all ten seeds with the rest of the pipeline built on it (Table 5). It is a worse emulator at every seed, 0.75% against 0.39% on the radiance, and its early stopping fired between 42 and 210 epochs of the 500 allowed: at this learning rate and batch size the wider network overfits the 18,884 rows before it fits them, and a wider network is not the way to close the network’s gap to the kernel on this table. Its representation is better. The exact kernel on its last-layer features reaches 0.079% on the radiance against 0.098% on the 512-unit features, below the per-input kernel’s 0.095% at all ten seeds and below the convex stack of Table 1 at nine of them, and on the two flux components, at 0.13% and 0.08%, it is at or below every row of that table. The convex stack over the wide pipeline’s heads, which puts 0.88 to 0.96 of its weight on that feature kernel, is the lowest row of the study at 0.078% on the radiance and 0.043 reflectance points, below the stack of Table 1 at nine of the ten seeds; the per-coordinate selection over the same heads does not transfer here either. The residual correction of the wider network is a little worse than that of the narrower one, as a correction of a worse predictor should be (Section 6). A

network	head	component rel. L^2 [%]				radiance	reflectance
		Y_1	Y_2	Y_3	Y_4	rel. L^2 [%]	med. $ \hat{\rho} - \rho $ [%]
3×512	network	0.48 ± 0.02	0.97 ± 0.04	0.41 ± 0.02	3.02 ± 0.18	0.387 ± 0.012	0.245 ± 0.008
	network + residual KRR	0.10 ± 0.00	0.30 ± 0.02	0.16 ± 0.01	2.43 ± 0.08	0.141 ± 0.006	0.089 ± 0.008
	kernel on its features	0.18 ± 0.01	0.17 ± 0.01	0.10 ± 0.01	1.99 ± 0.08	0.098 ± 0.007	0.054 ± 0.004
	per-coordinate selection	0.19 ± 0.01	0.44 ± 0.10	0.13 ± 0.02	2.25 ± 0.18	0.193 ± 0.022	0.123 ± 0.023
	convex stack	0.07 ± 0.00	0.16 ± 0.01	0.09 ± 0.00	1.94 ± 0.06	0.087 ± 0.006	0.048 ± 0.003
3×2000	network	0.93 ± 0.20	2.11 ± 0.49	0.76 ± 0.19	3.27 ± 0.19	0.748 ± 0.116	0.508 ± 0.110
	network + residual KRR	0.12 ± 0.01	0.35 ± 0.03	0.19 ± 0.03	2.35 ± 0.13	0.160 ± 0.016	0.106 ± 0.010
	kernel on its features	0.13 ± 0.02	0.13 ± 0.02	0.08 ± 0.01	2.10 ± 0.34	0.079 ± 0.009	0.045 ± 0.006
	per-coordinate selection	0.14 ± 0.02	0.30 ± 0.26	0.09 ± 0.01	2.55 ± 0.26	0.187 ± 0.057	0.099 ± 0.033
	convex stack	0.08 ± 0.01	0.13 ± 0.02	0.08 ± 0.00	2.00 ± 0.24	0.078 ± 0.011	0.043 ± 0.005

Table 5: The pipeline of Section 3 run with a 3×2000 network in place of the 3×512 one, at the same ten splits: the network (one member, up to 500 epochs under the same optimizer, learning rate and early stopping), its residual correction, the exact kernel on its last-layer features, and the per-coordinate selection and convex stack over the wide pipeline’s heads (the isotropic kernel, the network, its correction and its feature kernel), beside the corresponding rows of Table 1; mean $\pm$ standard deviation over the ten splits.

3×1000 rung at the same ten seeds sits between the two: the network at 0.72%, the kernel on its features at 0.088% against 0.098% at width 512 and 0.079% at width 2,000. From 512 to 2,000 the network’s own error rises with width while the error of a kernel on its features falls: what a wider network learns on this table is a better coordinate system, not a better map. At 4,000 units, run at two seeds, the network is worse again at 0.95% and the kernel on its features turns back up to 0.097%, so the gain in the representation does not continue past 2,000.

If that reading is right, several networks should give a better coordinate system than one. A three-member version of the same wide pipeline, run at three seeds, lets the exact kernel see the concatenated last-layer features of all three members at once, 6,000 coordinates rather than 2,000. It reads 0.074% on the radiance against 0.083% for the kernel on a single member’s features in the same lanes, lower at all three seeds, and 0.043 against 0.051 reflectance points, again at all three. That puts one exact solve level on the radiance with the convex stack of six heads fitted in those same lanes (0.074% against 0.074% on the mean, though the stack is the lower at two of the three seeds and on the reflectance median at all three). A combination of six predictors and a single kernel in a wider feature space buy about the same thing here, which is the plainest evidence in this study that what limits the kernel on this table is the coordinate system it is given and not the solve. Three seeds is an indication rather than a measurement, and the construction is not run at the width of Table 1, so whether the same gain survives there is left open.

5.1 Tuning with every row in the solve, and a metric read off the data

Two further constructions test the two readings above: that the bend at the top of the kernel’s curve is its tuning protocol, and that the metric the coordinate search finds is the physics of the components. Both were run at all ten seeds (Table 6).

Tuning with every row in the solve. The kernel families of Section 3 choose smoothness, scale and nugget by the validation error of a solve on a 6,000-row subsample, and refit the winner

kernel	component rel. L^2 [%]				radiance	reflectance
	Y_1	Y_2	Y_3	Y_4	rel. L^2 [%]	med. $ \hat{\rho} - \rho $ [%]
isotropic, tuned on 6,000 rows	0.08±0.01	0.42±0.03	0.20±0.00	2.62±0.18	0.239±0.016	0.141±0.018
isotropic, tuned on every row	0.08±0.00	0.36±0.02	0.17±0.01	2.58±0.14	0.210±0.008	0.107±0.006
per-input search, tuned on 6,000 rows	0.08±0.01	0.16±0.02	0.08±0.01	2.18±0.16	0.095±0.009	0.036±0.004
per-input search, re-tuned on every row	0.08±0.00	0.16±0.02	0.08±0.01	2.09±0.12	0.096±0.009	0.036±0.004
per-input from the cubic’s gradients, $p = 1$	0.34±0.01	0.19±0.02	0.14±0.01	2.17±0.14	0.159±0.007	0.078±0.004
per-input from the cubic’s gradients, $p = 1/2$	0.11±0.00	0.19±0.02	0.09±0.01	2.27±0.15	0.125±0.008	0.055±0.004
per-input from the cubic’s gradients, $p = 1/4$	0.08±0.00	0.25±0.02	0.14±0.01	2.38±0.20	0.157±0.015	0.077±0.013

Table 6: Where the kernel’s hyperparameters are chosen, and how its metric is obtained, at the full training block and the same ten splits. The first pair tunes smoothness, scale and nugget on a 6,000-row subsample and on every training row; the last three rows replace the coordinate search by a metric read off the cubic surrogate’s gradients, at three exponents. Selecting the exponent on validation, per seed, chooses $p = 1/2$ at all ten, so that row would duplicate the $p = 1/2$ row and is not printed; the per-seed choices are in the stored numbers. Mean $\pm$ standard deviation over the ten splits.

on all 18,884 rows. Doing the tuning solve on every row instead lowers the isotropic kernel’s radiance error from 0.239% to 0.210% and its reflectance median from 0.142 to 0.107 points, better at nine of the ten splits, the ratio averaging 0.882 and never above 1.001. The mechanism is visible in the chosen scales: over the forty component-seed cells the full block picks a shorter length scale than the subsample in 26, the same in 12, and a longer one in 2. A tuning set a third of the size of the solve supports a longer length scale than the solve does, so the subsample protocol over-smooths, and it over-smooths more the larger the gap between the two.

That is the bend. Below 6,000 rows the tuning subsample *is* the whole training block and the rungs are already full-block tuned; at 8,000 rows it is three quarters of it; only the top rung is a third. Applying the measured ratio to the top rung of Table 2 takes the kernel’s last step on the radiance from -0.183 to -0.334 , against a slope of -0.305 over the five lower rungs: the flattening at the top disappears, and the curve becomes a straight line in $\log n$ over a factor of 38 in data. On the two flux components the same correction takes the last step from -0.187 and -0.171 to -0.352 and -0.358 against averages of -0.437 and -0.468 , so there most of the bend is the protocol and a little of it is not. The consequence for the comparison of Section 4 is that the kernel’s margin over the network at the full block is wider than Table 2 shows: 1.8 times rather than 1.6, and the kernel is not slowing down at the sizes this table reaches.

The same re-tuning does nothing for the per-input kernel: 0.096% against 0.095%, better at four of the ten splits, ratio 1.005. Its coordinate search already selects a scale of once the median distance at the metric it finds, which is what the full-block tune selects too. Rescaling the coordinates removes the over-smoothing that the isotropic kernel suffers, so the per-input kernel of Table 1 was never paying the tuning cost, and its number stands as reported.

A metric read off the data. The coordinate search costs a sweep over the six inputs at five multipliers, twice, per component and per seed. The alternative sets the metric before any kernel is fitted: the multiplier of coordinate j is the root-mean-square gradient of the fitted cubic surrogate along j over the training rows, raised to a power p , normalized to geometric mean one and clipped to the search’s own range $[1/16, 4]$, so a coordinate the map barely responds to gets a long length scale and p sets how hard the metric leans on that reading. It costs one cubic fit and six finite differences.

Taken proportionally, at $p = 1$, it captures rather more than half the gap: on the radiance it reads 0.159%, below the isotropic kernel at every one of the ten splits by a factor 0.669 on average and above the searched metric at every one of them by a factor 1.680. But p is a free number, and the ladder is not monotone. The square root reads 0.125% and the fourth root 0.157%, so the best exponent is interior, and choosing it the way every other hyperparameter in this study is chosen — the lowest mean validation error over the four components, per seed, with the test block untouched — selects $p = 1/2$ at all ten seeds. At that exponent the metric closes 79% of the distance from the isotropic kernel to the searched one on the radiance and 82% on the reflectance median, for one cubic fit, six finite differences and a choice among three numbers.

The per-band reading says why the exponent matters and what the search still buys. The four components do not want the same one: Y_2 and Y_3 are best at $1/2$, the albedo at 1 (2.18% against 2.27%), and the path radiance at $1/4$. Y_1 is where the damage is, and it is worth seeing why. On that component the isotropic kernel already reads 0.084% and the searched metric 0.085%: there is no anisotropy to find. The proportional metric nonetheless reads 0.34% there, four times worse than doing nothing, because the surrogate’s gradient along a coordinate measures how fast the map moves with it and not how much predictive information it carries, and on the one component that depends on the viewing geometry those two disagree. Softening the exponent to $1/4$ takes Y_1 back to 0.085%, exactly the searched metric’s value, at the cost of Y_2 and Y_3 . So a metric read from first-order sensitivity, with its exponent validated like any other hyperparameter, is most of what the coordinate search buys at a small fraction of its cost. What the search still buys is the freedom to be anisotropic differently for each component, which one global exponent cannot be.

6 What the residual correction inherits

The curves of Section 5 show a corrected network whose error decays at the network’s exponent and sits at the kernel’s level, below both parents on the flux components and the radiance from a few thousand rows on, and above the kernel alone on the path radiance at every size. One identity accounts for all of it.

Fix a training design $X_n = \{x_1, \dots, x_n\}$, a kernel k with Gram matrix K on the design, and a nugget $\lambda > 0$. Kernel ridge regression maps a function g , known on the design, to

$$\hat{g}(x) = k(x, X_n) (K + \lambda I)^{-1} g(X_n), \quad (4)$$

and this map is linear in g . Write $Rg = g - \hat{g}$ for the residual of the regression. For a target f and a trained network h , the residual-corrected predictor of Section 3 is $\hat{f}_{\text{corr}} = h + \widehat{(f - h)}$. The statements below hold coordinatewise for vector-valued f and h , so one coordinate is written.

Proposition 1. (i) $f - \hat{f}_{\text{corr}} = R(f - h)$: the error of the corrected predictor is the kernel regression’s residual on the network’s residual. (ii) On the design, $Rg(X_n) = \lambda (K + \lambda I)^{-1} g(X_n)$. In the eigenbasis $K = \sum_j \mu_j u_j u_j^\top$ the coefficient of $g(X_n)$ along u_j is multiplied by $\lambda/(\mu_j + \lambda) \in (0, 1)$: the correction removes from the network’s training residual its projection on the leading eigenvectors of the Gram matrix and leaves the trailing ones. (iii) Suppose the regression satisfies $\|Rg\|_{L^2(P)} \leq C_n \|g\|_{\mathcal{S}}$ for every g in a class $\mathcal{S}$ with norm $\|\cdot\|_{\mathcal{S}}$, and $f, h \in \mathcal{S}$. Then

$$\|f - \hat{f}_{\text{corr}}\|_{L^2(P)} \leq C_n \|f - h\|_{\mathcal{S}} \leq C_n (\|f\|_{\mathcal{S}} + \|h\|_{\mathcal{S}}), \quad (5)$$

and the corrected predictor is inside the kernel's own bound $C_n\|f\|_S$ exactly when $\|f - h\|_S < \|f\|_S$.

Proof. (i) By linearity of (4), $\widehat{(f - h)} = \hat{f} - \hat{h}$, so $f - \hat{f}_{\text{corr}} = (f - \hat{f}) - (h - \hat{h}) = Rf - Rh = R(f - h)$. (ii) On the design $\hat{g}(X_n) = K(K + \lambda I)^{-1}g(X_n)$, so $Rg(X_n) = (I - K(K + \lambda I)^{-1})g(X_n) = \lambda(K + \lambda I)^{-1}g(X_n)$; the eigen-expansion of $(K + \lambda I)^{-1}$ gives the factors. (iii) Apply the bound to $g = f - h$ and the triangle inequality. $\square$

The proposition is algebraic, so what needs checking is that the implementation is the object it describes: that the correction enters linearly and that no centring or re-standardization between the stages breaks that. A script accompanying the records builds a small design, runs the campaign's own kernel and solver, and verifies (i) to machine precision, the linearity it rests on, the design identity of (ii) with its shrinkage factors, that every factor lies in $(0, 1)$ and is smallest on the leading eigendirections, and the design-space form of (iii) with the sharp constant $\lambda/(\mu_{\min} + \lambda)$. It also reproduces the failure mode: when the network's residual is made rougher than the target, correcting the network scores worse than regressing the target directly, which is what Y_1 does in Table 1, where the corrected network reads 0.10% against the isotropic kernel's 0.08%.

Bounds of the form assumed in (iii) are the standard learning rates for kernel ridge regression: for g in an interpolation space between $L^2(P)$ and the reproducing kernel Hilbert space of k , with the nugget chosen on the right schedule, C_n decays as a power of n whose exponent depends on the source condition and on the eigenvalue decay of the kernel and not on $\|g\|_S$, which enters as a constant [16, 17]. What (iii) then says is that the corrected predictor inherits the kernel's exponent with the residual's norm as its constant. The network's training drives $\|f - h_n\|_S$ down as n grows, so the corrected curve falls at least as fast as the kernel's bound and, through the constant, tracks the network: its exponent is the network's, its level the kernel's, which is what Table 2 measures on the radiance and Table 3 and Figure 3 on the two flux components. The bound also allows the exception. On the path radiance the isotropic kernel alone beats the correction at every training size, and (iii) puts the corrected error at $C_n\|f - h\|_S$ against the kernel's $C_n\|f\|_S$: when the network's residual is rougher, in the norm the kernel measures, than the smooth map it approximates, correcting the network is worse than regressing the map. Part (ii) is the mechanism of Section 4 stated exactly: the correction hands the network the kernel's accuracy on the smooth directions of its residual, which are the leading eigenvectors of the Gram matrix, and leaves what is rough. It is also why the correction is the construction that does not transfer to a rough map (Section 7), where the residual has little weight on those directions, and why its optimum in Table 4 lies at a moderate rank, where the trailing coefficients it is asked to regress are the network's noise rather than the map's tail.

7 Where the constructions do not transfer: OCO-2

The companion paper [6] applies the same construction to the OCO-2 radiative-transfer emulation problem of Lamminpää et al. [5]: per spectral band (O_2 , weak CO_2 , strong CO_2), a reduced atmospheric state of twenty or twenty-four coordinates maps to forty reduced radiance coefficients, with errors on the standardized coefficients and on the reconstructed radiance, under the ten-split protocol of Section 2. That paper carries the tables; the findings that are the counterpart of the EMIT results are stated here so the two problems can be read against each other.

The problem sits at the opposite end of the structural spectrum. The input is higher-dimensional and the map rougher, and a kernel on the state trails the network by a factor of nine, four and five on the reduced coefficients: the isotropic Matérn reaches 40.6%, 58.6% and 43.4% on the three bands against 4.4%, 16.4% and 8.2% for the network. The residual correction, the EMIT table’s second-best construction, adds nothing on any band: on a rough map the residual of a trained network carries little weight on the leading eigenvectors of the Gram matrix, and Proposition 1(ii) leaves it as it is. The coordinate search that gives a kernel on the state its own length scales, the strongest construction on the EMIT table, is not the strongest there either. Four learned metrics were run at the same ten splits on all three bands: a kernel-flow metric [26], a Mahalanobis metric under the same flow, the gradient outer product of the recursive feature machine [15], and per-input scales fitted by empirical Bayes. The last is the best of the four at every band, taking the isotropic kernel’s 40.6%, 58.6% and 44.3% on the coefficients to 18.7%, 25.3% and 16.9%, and on O_2 its radiance from 0.233% to 0.150%; the two flow metrics agree within half a point of each other at 26.9%, 29.8% and 23.9%, the recursive feature machine reads 28.2%, 47.9% and 31.3%, and an additive kernel 38.2%, 46.9% and 47.7%. The isotropic baseline agrees between these lanes and the protocol lanes on O_2 and weak CO_2 to two decimals; on strong CO_2 it reads 44.3% here and 43.4% there, the one band where they part, and also the one whose isotropic column carries a spread of ± 3.4 because two of its ten splits land three points below the other eight. So a learned metric does close a good part of the distance on the state, cutting the isotropic error to between two fifths and a half of itself depending on the band. What it does not reach is the kernel inside the representation, and by how much is a property of the band rather than a constant: the exact Matérn regression on the network’s frozen features reads 4.1%, 16.3% and 8.1% at the same splits, so the best state metric is 4.6 times that on O_2 but only 1.6 and 2.1 times it on the two CO_2 bands. One caution on the best row: on O_2 the empirical-Bayes metric is bimodal across splits, eight of the ten landing near 17.2% and two at 24.7%, so its spread of ± 3.2 is thirty times the isotropic kernel’s ± 0.1 and its mean is not the value a single split is likely to show. That last construction is the one thing that does transfer, and it transfers with room to spare: the exact Matérn regression on the network’s frozen features beats a ridge readout refitted on the same features at every split, on every band and under every training loss, and a per-coefficient choice among heads trained under different losses holds both reported metrics at once, below the release’s stored emulator on both; those results and their tables are in [6].

One observation about the release itself is recorded here because it does not depend on the protocol. The reference’s data release carries 976 retrieval records, each with an atmospheric state, the full-physics monochromatic radiance at that state, its rank-40 reconstruction, and the prediction of the release’s Gaussian-process emulator at the same forty coefficients. Every one of the 976 reduced states matches a stored test input within 6.67×10^{-16} ; the records map to 974 distinct test rows, and the nearest test-row index is the same in all three bands. The records are therefore not disjoint from the stored test set and cannot serve as an independent evaluation pool, though they do provide a different target, the full-physics radiance, at those states. The state reduction was replayed independently from the release’s two dimensional-reduction files and the `x_in` and `geo_in` fields of all 976 records; the replay code and its digests are archived with the per-run records.

8 Ten further emulation configurations

The two problems above stand at opposite ends of one axis: a smooth, six-dimensional, low-rank table where an exact kernel with a learned metric is the strongest single model, and a rougher, twenty-dimensional one where the kernel is useful only inside a learned representation. If that axis orders the comparison, it should order other emulation problems too. Table 7 applies the same family set to ten further configurations, with no tuning per corpus beyond what each family selects on validation.

The corpora are the advection equation with discontinuous initial data from the operator suite of [18], which is also the source of the structural-mechanics problem of [6]; Burgers’ equation at three viscosities from PDEBench [19]; three one-step maps from The Well [20] — a two-dimensional turbulent radiative layer, an active-matter system, and a Helmholtz staircase; the ClimSim atmospheric physics emulation task [21]; and the paired correction-coefficient corpus of the physics-guided radiative-transfer work of [4], which is a third radiative-transfer emulation problem and the largest here at 487,790 rows. That last corpus is run in both of its configurations: from the atmospheric state alone, and with the low-fidelity 6S coefficients supplied as additional inputs, which is a different and easier problem.

The configurations are these. Advection is the 200-point input/output pair of the operator suite, used at full resolution with no reduction. Burgers is PDEBench’s `1D_Burgers_Sols_Nu/` release, the map from the initial condition to the solution at $t = 1$ on a grid subsampled by four to 256 cells, the input used raw and the output reduced to 128 principal coefficients. The three Well datasets are one-step maps of the full multi-channel state with 256 principal coefficients in and out: the turbulent radiative layer on its 128×384 grid (density, pressure, velocity), active matter on 256×256 (concentration, velocity, and the two tensor fields), and the Helmholtz staircase on its two pressure channels. ClimSim is the low-resolution release, 124 inputs to 128 tendencies, with 100,000 training rows drawn from the 10.1M available and validation and test taken from the release’s own validation and scoring files. The correction-coefficient corpus is the paired 6S/libRadtran Sentinel-2 set, nine state variables and the band wavelength mapping to the three libRadtran coefficients, split by atmospheric state so that all bands of a state travel together; its multi-fidelity configuration adds the 6S coefficients of the same state as inputs.

The families are those of Section 3 with two differences forced by scale. The reference family is a ridge regression linear in whatever inputs the models receive, not the cubic of Table 1: a cubic expansion of a 256-dimensional input is not tractable, and on the two corpora whose inputs are six- and nine-dimensional the linear form is the one that was run. And the kernel solves are exact only when the training block has at most 20,000 rows; above that they use a Nyström approximation on 6,000 uniformly drawn landmarks. That cap binds on ClimSim, on the Helmholtz staircase at 20,384 rows, and on the two correction-coefficient configurations. Those entries are what a capped fit gives; they are not the exact solve, and we do not claim them as a bound on it in either direction, since restricting the fit set both discards information and regularizes. The convex stack and the per-coordinate selection are formed from the kernels, the networks, the residual correction and the feature kernel; the ridge is a reference row and is not one of their members.

What each row averages differs and is stated in the table. Advection, Burgers, ClimSim and the two correction-coefficient configurations have three to five random splits. The Well’s releases carry their own train, validation and test files, which are used unchanged, so the three turbulent-radiative-layer runs share one split and differ only in network initialisation; their spread, of order 0.01 points, measures initialisation and not sampling. The active-matter and

corpus	n	d	q	runs	ridge (linear)	KRR	KRR, learned metric	network	network ensemble	network + residual KRR	kernel on features	convex stack
Advection	19,000	200	200	5	11.29±0.04	11.36±0.04	11.36±0.04	11.69±0.07	11.50±0.05	11.50±0.05	11.64±0.07	11.36±0.04
Burgers $\nu = 0.1$	7,200	256	128	3	13.89±0.53	2.81±0.28	2.90±0.33	20.34±0.61	19.28±0.09	2.78±0.29	4.92±0.30	2.78±0.29
Burgers $\nu = 0.01$	7,200	256	128	3	33.68±0.28	23.40±0.54	23.58±0.50	36.65±1.95	35.31±0.70	23.12±0.56	29.20±0.59	23.01±0.54
Burgers $\nu = 0.001$	7,200	256	128	3	38.38±0.12	31.66±0.16	33.23±1.73	39.98±0.65	38.89±0.11	31.82±0.11	35.59±0.48	31.34±0.14
Turbulent radiative layer	7,200	256	256	3	31.73	30.31	30.26±0.01	32.17±0.02	31.58±0.02	30.40±0.01	31.08±0.01	30.12
Active matter	14,000	256	256	1	37.24	35.81	35.18	37.31	36.93	35.77	36.22	35.05
Helmholtz staircase	20,384	256	256	1	0.00	0.26	0.07	0.40	0.24	0.23	0.45	0.07
ClimSim	100,000	124	128	5	32.24±0.06	27.20±0.05	26.31±0.04	25.10±0.18	24.43±0.17	24.33±0.13	24.71±0.25	24.25±0.17
RT corrections, state inputs	487,773–487,823	8	3	5	71.54±0.59	62.93±0.51	3.97±0.42	3.75±0.25	3.24±0.11	3.13±0.05	3.52±0.07	3.09±0.05
RT corrections, multi-fidelity inputs	487,773–487,823	11	3	5	33.83±0.26	15.76±1.98	4.60±0.31	2.79±0.14	2.40±0.11	2.34±0.09	2.80±0.09	2.32±0.08

Table 7: Mean relative L^2 test error [%] on ten further emulation configurations, mean $\pm$ standard deviation over the runs available for each. n is the number of examples — a range where the state-level split of the correction-coefficient corpus makes it differ by run — d the input dimension the models receive and q the output dimension they regress. The runs are random splits except for the turbulent radiative layer, whose three runs share The Well’s own split and differ in network initialisation, and the two single-run rows. The family set and the validation-based selection are common to every row; the network budget is not, and is 200 epochs at width 512 except on ClimSim (60 at 512) and the two correction-coefficient configurations (120 at 384). The ridge column is linear in the inputs the models receive, unlike the cubic reference of Table 1, and is not a member of the convex stack; its Helmholtz-staircase entry is 0.004, below the two decimals shown. Kernel solves are exact when the training block has at most 20,000 rows and otherwise use 6,000 Nyström landmarks, which is the case on ClimSim, the Helmholtz staircase and the two correction-coefficient rows.

Helmholtz-staircase rows are single runs.

Three things in the table repeat what the two main problems show. First, the ordering between families follows the structure rather than the corpus. Where the map is smooth and the sample small — Burgers at $\nu = 0.1$ — the exact kernel is at 2.81% against the network’s 20.34%, a factor of seven, of the same kind as the five- to six-fold gap between the per-input kernel and the network on the EMIT table’s smooth components; as the viscosity falls and the solutions roughen the gap closes to 23.40% against 36.65% and then to 31.66% against 39.98%. Where the sample is large and the map rough — ClimSim, and the correction-coefficient corpus at nearly half a million rows — the network leads instead, 25.10% against 27.20% and 3.75% against 62.93%, which is the OCO-2 regime. Where the operator is linear, on advection, the linear ridge is the lowest row and every other family lands within 0.4 points of it, as it should.

Second, a learned metric is what rescues a kernel on the state when the plain one fails, and it is worth most where the plain kernel is worst. On the correction-coefficient corpus from the state alone the isotropic kernel is at 62.93 ± 0.51 and a kernel-flow metric [26] takes it to 3.97 ± 0.42 , a factor of sixteen in the mean, which brings a kernel on the state to within six percent of the network on a corpus where the plain kernel was seventeen times worse. With the low-fidelity coefficients supplied as inputs the same pair reads 15.76 and 4.60, a factor of three: the easier the problem is made for the plain kernel, the less the learned metric adds. This is the same mechanism as the per-input length scales on the EMIT table and the empirical-Bayes metric of Section 7, at a scale where its effect is unambiguous.

Third, the combination is at or near the best row of its own pool everywhere, and where it is not the lowest row of the table the reason is membership rather than weighting. The convex stack is the lowest row on eight of the ten configurations and the second lowest on the other two; the residual-corrected network is in the lowest two on five. Both exceptions are the linear problems, and on both the row that beats the stack is the ridge, which the stack does not contain. On advection the ridge is lower by 0.07 points while the stack, at 11.36%, matches the best member it has — the isotropic kernel, on which it puts weight 0.87 — to two decimals.

On the Helmholtz staircase the ridge reaches 0.004%, sixteen times below the next row of the table and about a hundred times below the weakest member of the pool, and the stack reaches 0.066%, again a shade below its own best member, the learned-metric kernel at 0.067%, on which it puts weight 0.94. In both cases the stack does what it is asked to do inside its pool; what it cannot do is recover a family the pool excludes. The statement this supports is correspondingly narrow: a combination is a good default when no member of its own pool dominates it, and a combination can only be read against the whole comparison when the pool contains the families the comparison includes.

Three limits on scope. These rows use one training budget per corpus and one architecture family, and several of these corpora have published results from methods tuned for them specifically; the comparison here is between families under a common protocol, not against the state of the art on each problem. With three to five runs, and one on two rows, the spreads support the large orderings above and not fine distinctions between adjacent families. And every kernel column here inherits the tuning protocol Section 5.1 measures: the length scale is chosen on a 6,000-row subsample and the winner refit on every row, which over-smooths by an amount that grows with the ratio of the two. On the EMIT table that costs the isotropic kernel a factor 0.882; on ClimSim, where the subsample is a sixteenth of the solve, re-choosing the scale against the solve that actually runs halves it and lowers the kernel’s error by 3.6%; on a 20,000-row corpus it changes nothing. The kernel columns of Table 7 are therefore pessimistic by an amount that differs by row and is not measured here, and the gaps they show to the network columns are upper bounds on the true ones.

9 Discussion

The EMIT comparison settles three things, and on each of them the ten-split protocol changes what the single split reported. The first is the standing of the exact kernel. Fitted on a 4,000-point subset with a single length scale it is a useful baseline; fitted on every training row with a length scale per input it is the best single model on the EMIT table, and its metric is interpretable, since the coordinate search finds, from the validation rows alone, that the transmittances and the spherical albedo do not depend on the relative azimuth. The second is the standing of the corrected network. Its mechanism is the one described in [6] — the exact regression of the residuals hands the network the kernel’s accuracy on the smooth components while the network keeps its own representation — but with the kernel allowed its own metric the correction is one of four heads within a few hundredths of a percent of one another on the radiance, not the lowest row. The third is the way heads are combined. A per-coordinate selection made on 2,098 validation rows does not transfer to the test block, and its spread across splits is among the largest in the table; a convex stack per component, fitted in the reported metric, is the best row at every split but one. We read this as a statement about calibration data rather than about the heads: with a few thousand rows the right way to use complementary predictors is a low-dimensional combination, not a per-output choice.

The curves of Section 5 say how far each of those statements carries. The cubic’s floor is reached by a thousand rows; every other family falls at close to $n^{-1/2}$ on the smooth components and not at all on the albedo, so the table’s ordering is an ordering of constants and not of rates, and the kernel’s margin over the network, 3.1 times at 500 rows and 1.6 times at the full block, is a margin at this training size. The bend at the top of the kernel’s curve is its tuning protocol and nothing else: tuning with every training row in the solve, rather than a 6,000-row subsample,

lowers the isotropic kernel’s radiance error by a factor 0.885 there, straightens the curve to a line in $\log n$ over a factor of 38 in data, and widens the kernel’s margin at the full block from 1.6 to 1.8; the per-input kernel, whose search rescales the coordinates first, never pays that cost. A tuning set smaller than the solve supports a longer length scale than the solve does, and it over-smooths by exactly that much. The identity of Section 6 then accounts for the corrected network’s curve without a free parameter: its error is the kernel regression’s residual on the network’s residual, so it inherits the kernel’s rate with the residual’s norm as its constant, which is why it takes the network’s exponent, sits at the kernel’s level, lies below both parents on the flux components from a few thousand rows on, and stays above the kernel alone on the path radiance, where the network’s residual is rougher than the map. The same identity, read on the training design, is the mechanism the single split reported: the correction removes the projection of the network’s residual on the leading eigenvectors of the Gram matrix and leaves the rest, which is why it does not transfer to a rough map and why it has an optimum in the output rank where the kernel itself has none.

Width changes the network’s standing in one direction and its representation’s in the other. A network four times wider is a worse emulator on this table, stopping early at every seed, and its last layer is a better coordinate system for the kernel than the narrower network’s: the exact kernel on those features beats the per-input kernel at every seed, and the convex stack built on the wide pipeline is the lowest row of the study. What a wider network learns here is not a better map but a better metric, and the exact solve is what turns the metric into accuracy.

The OCO-2 problem of the companion paper says the same thing from the opposite structure. There the kernel on the state is four to nine times behind the network, the residual correction adds nothing, the learned metrics on the state cut its error roughly in half but leave it between one and a half and five times the kernel inside the representation, which is the construction that helps (Section 7). Section 8 applies the same family set to ten further configurations, where the family ordering follows the structure as it does here, a learned metric rescues a kernel on the state exactly where the plain kernel fails worst, and the stack is the lowest row on eight of the ten and the second lowest on the other two, in both cases behind a reference family its pool does not contain.

The spherical albedo is the hard component throughout. Every family in Table 1 is an order of magnitude worse on Y_4 than on the other three, and the per-band anatomy places that error where the albedo is close to zero, so relative error is large there for any predictor. No family in this pool moves it, at any training size in Table 2, at any output rank, or at four times the width. An albedo error reaches an emulator used through the inversion (2) directly and everywhere, not only in the cancellation tail: by (3), $\partial\hat{\rho}/\partial\hat{Y}_4 = -\hat{\rho}^2$, so at $\rho = 0.7$ an albedo error moves the recovered reflectance by about half its own size at every band, whatever the transmitted flux. What the tail of Section 2 amplifies is the other three components, whose sensitivities carry the vanishing denominator.

On cost, the exact solve is one Cholesky factorization of an $18,884 \times 18,884$ matrix per component and one kernel row per query; the per-input search multiplies the tuning cost by the number of coordinates and leaves the solve unchanged. The networks cost more to train and less to query. The stack costs nothing beyond its members.

Only differences that exceed the seed-to-seed spread should be read as comparisons.

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Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author made substantial use of generative AI tools, and describes their role here. The software — the data pipeline, the model comparison, the diagnostics, and the figures — was written and executed by Claude Code (Anthropic) and Codex (OpenAI), which also prepared the first draft of this manuscript, working from the author’s data and from the method and experimental recipe of [6]. The measurements of the data’s structure, the identification of the ill-posed reflectance entries, the scaling runs of Section 5 and the identity check of Section 6 were produced in the same AI-assisted sessions and verified against the run summaries. The author reviewed the code, the numbers, and the text, and takes full responsibility for the content of the manuscript.

Data and code availability

The dataset (`data_EMIT_24k`) consists of tabulated radiative-transfer evaluations generated at the Jet Propulsion Laboratory; the OCO-2 data and the reference emulator’s stored predictions are those of the companion paper’s release; the corpora of Section 8 are public releases, cited there. The pipeline, the per-seed records behind every number reported here (one record per seed for the EMIT table and one per seed and rung, rank, width or metric exponent for Section 5, each with its test predictions), and the exact configuration of each experiment are retained by the author and will accompany the published version of this manuscript; the method repository of [6] is at <https://github.com/yspennstate/neural-means-kernel-corrections>.

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